

Part C: BIOS vs UEFI

Criteria	BIOS (Basic Input/Output System)	UEFI (Unified Extensible Firmware Interface)
Definition	Legacy firmware stored on a ROM/flash chip that initializes hardware during startup and loads the OS.	Modern firmware interface designed to replace BIOS, offering a flexible software environment and advanced hardware control.
Boot Process	Executes POST, reads the Master Boot Record (MBR) on sector 0, and hands execution to the bootloader stage.	Initializes hardware independently, reads the EFI System Partition (ESP), and launches OS bootloaders directly via <code>.efi</code> executables.
Maximum Disk Support	Limited to 2.2 TB per drive due to 32-bit logical block addressing.	Supports up to 9.4 ZB (Zettabytes) with 64-bit addressing.
Partition Style	Master Boot Record (MBR) — maximum of 4 primary partitions (or 3 primary + 1 extended).	GUID Partition Table (GPT) — supports up to 128 partitions in standard setups without extended partitioning.
Security Features	Basic supervisor/user boot passwords; vulnerable to rootkits and boot sector malware.	Advanced security features including Secure Boot, cryptographic signature validation, and hardware-backed TPM integration.
Speed	Slower boot times due to 16-bit processor mode execution and sequential device initialization.	Significantly faster initialization via 32-bit/64-bit mode, parallel hardware initialization, and direct EFI loading.
Advantages	Simple, lightweight architecture; universally supported by legacy 16-bit and 32-bit operating systems.	Large storage capacity support, faster initialization, graphical mouse-driven setup UI, network booting out-of-the-box, and superior security.

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Disadvantages	Hard 2.2 TB storage limitation, slow execution, poor security against low-level malware, limited to 1 MB memory space.	Higher complexity, larger firmware size, potential compatibility issues with legacy operating systems without CSM mode enabled.
Modern Usage	Largely deprecated; used mainly in legacy embedded systems or through UEFI's Compatibility Support Module (CSM).	Standard industry firmware for all modern desktop PCs, laptops, enterprise servers, and cloud virtualization platforms.